# Sarang P Karun

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## Professional Summary

AI/ML engineer with hands-on experience building and deploying LLM applications, computer vision systems, data pipelines, and automation workflows. Backed by a research background in deep learning, with a focus on delivering practical, production-ready AI solutions.

## Experience

**AI/ML Development Engineer** May 2025 – Present
*ThinkNeural AI - Ernakulam, Kerala*

- Architected **Datadime**, a 14-agent business intelligence platform (FastAPI, DuckDB) that answers natural-language questions using LLM tool-use orchestration, sandboxed SQL generation, semantic search (Qdrant), forecasting, and auto-generated charts.
- Built a role-based, multi-provider LLM layer (22 roles across multiple models, including self-hosted vLLM), and used **Langfuse** tracing and eval harnesses to test cheaper models before routing.
- Made the orchestrator report data gaps instead of fabricating answers, and shipped an embeddable FAQ Bot widget.
- Built real-time video analytics for **Sakshi AI** using YOLO, NVIDIA DeepStream, and custom segmentation across 100+ camera feeds, deployed with FastAPI and Docker.
- Developed **annotation automation** for a computer vision training platform, connected to a CI/CD retraining pipeline to cut manual labeling and speed up model iteration.

**Junior Data Analytics Intern** December 2025 – April 2026
*Knovista Technologies Pvt. Ltd. — Ernakulam, Kerala*

- Built an interactive Power BI sales dashboard covering profitability, customer behavior, and regional performance, with Power Query workflows and DAX measures for key KPIs.

**Junior Research Fellow** April 2025 – November 2025
*Rajagiri School of Engineering & Technology — Ernakulam, Kerala*

- Developed deep learning models for flood susceptibility prediction from remote sensing data, with an interactive risk dashboard built on Python, TensorFlow, and Google Earth Engine.

**Junior Research Fellow** August 2024 – March 2025
*National Institute of Technology — Calicut, Kerala*

- Developed KNN-based models (scikit-learn) for route safety and dynamic risk prediction.
- Built REST APIs and cloud services with Python, Flask, GCP, and Firebase, integrating real-time data and Generative AI.

## Projects

**Cold Email Automation** | *Flask, SMTP, Python*

- Built an end-to-end cold outreach platform for Sakshi AI with CSV prospect import, a manual review and approval UI, and an optional auto-send mode over SMTP.
- Used a self-hosted LLM (Qwen2.5-14B via vLLM) to write personalized emails for each prospect from scraped brand information, avoiding third-party API costs and keeping prospect data in-house.

**Multimodal Sarcasm Detection** | *Python, PyTorch, ALBEF*

- Built an ALBEF text-and-image sarcasm detector (87.95% accuracy) and published a paper.

## Technical Skills

**Languages**: Python, SQL, JavaScript
**AI/ML & LLMs**: LLMs, Agentic AI, RAG, Prompt Engineering, PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, Langfuse, vLLM, Ollama, Qdrant
**Computer Vision**: YOLO, OpenCV, NVIDIA DeepStream, Image Segmentation
**Backend & Data**: FastAPI, Flask, REST APIs, PostgreSQL, DuckDB, Pandas, NumPy, React.js, Power BI, DAX
**DevOps & Cloud**: Docker, Git, CI/CD, Google Cloud (Vertex AI, Firebase)

## Education

**Central University of Kerala** Kasaragod, India
*Master of Science — Computer Science (Intelligent Systems), CGPA: 9.00* *July 2024*
**NSS College of Engineering** Palakkad, India
*Bachelor of Technology — Computer Science Engineering, CGPA: 8.47* *June 2022*